Gold Mining & Exploration

A Complete Guide

Internal Training Material

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Classification: Internal Training Document

GOLD MINING TRAINING GUIDE

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PART	TOPIC	ALL	LEGAL	FINANCE	BD	OPS	MGMT
1	**Gold Fundamentals**	●	●	●	●	●	●
2	**Deposit Types**	●	◐	◐	●	●	●

PART	TOPIC	ALL	LEGAL	FINANCE	BD	OPS	MGMT
3	**Exploration**	●	◐	●	●	●	●
4	**Resource & Reporting**	●	●	●	●	◐	●
5	**Mining & Processing**	◐	○	◐	◐	●	◐
6	**Economics & Valuation**	◐	◐	●	●	◐	●
7	**Legal & Regulatory**	◐	●	◐	●	◐	●
8	**Value Chain & Market**	◐	◐	●	●	○	●
9	**Industry Landscape**	○	○	◐	●	○	●
10	**Glossary**	Reference — Use as needed					

● Essential ◐ Recommended ○ Optional

Part 1: Gold Fundamentals ALL

1.1 Why Gold Matters

Gold (chemical symbol: **Au**, from Latin *aurum*) has been valued by humans for over 5,000 years. Understanding its basic physical properties is essential because they directly determine how gold deposits form, how we find gold, and how we extract it.

PROPERTY	VALUE	WHY IT MATTERS
Density	19.3 g/cm³	~7× heavier than rock. This is why gold sinks to the bottom of rivers and why gravity separation works
Chemical Inertness	Does not rust or corrode	Gold survives millions of years of weathering. The rock disappears, but the gold remains
Malleability	Most malleable metal	Gold deforms rather than breaks during river transport — flattened into flakes and nuggets
Melting Point	1,064°C	Achievable with basic equipment, enabling on-site smelting
Crustal Abundance	0.004 ppm	Extremely rare. Even a 1 g/t deposit represents 250× natural concentration

> **Practical implications:** Gold's extreme density is the foundation of alluvial mining operations — gravity methods work because gold is so much heavier than everything else. Gold's chemical inertness means it survives weathering and accumulates in rivers over millions of years, creating the alluvial deposits found across West Africa.

1.2 How Gold Is Measured

Grade

Grade is the concentration of gold in rock or sediment — **the single most important number in gold mining**.

UNIT	MEANING	CONTEXT
g/t (grams per tonne)	Grams of gold per tonne of rock	Standard for hard rock mining. 1 g/t = 1 ppm
g/m³ (grams per cubic metre)	Grams of gold per cubic metre of gravel	Standard for alluvial mining
oz/t (ounces per short ton)	Troy ounces per short ton	Older North American reports. 1 oz/t = 34.29 g/t

Grade Benchmarks — Hard Rock

CATEGORY	GRADE RANGE
Earth's average crustal gold	0.004 g/t (background)
Low grade	0.5–1.0 g/t
Medium grade	1.0–5.0 g/t
High grade	5.0–20 g/t
Bonanza grade	>20 g/t

Grade Benchmarks — Alluvial

CATEGORY	GRADE RANGE
Sub-economic	<0.1 g/m³
Low grade	0.1–0.3 g/m³
Medium grade	0.3–0.5 g/m³
Good grade	0.5–1.0 g/m³
High grade	1.0–2.0 g/m³
Very high grade	>2.0 g/m³

Key Unit Conversions

CONVERSION	VALUE
1 troy ounce (oz)	31.1035 grams
1 metric tonne	1,000 kilograms
1 kilogram of gold	32.15 troy ounces
1 short ton	907.185 kilograms

Purity (Fineness)

FINENESS	PURITY	CONTEXT
999.9	99.99% — "four nines"	Investment grade, highest purity
995	99.5%	LBMA Good Delivery minimum
900	90%	Typical doré bar from alluvial operations
750	75% (18 karat)	Common jewellery standard

Part 2: Types of Gold Deposits ALL

2.1 Two Worlds: Hard Rock vs Alluvial

Primary (Hard Rock / Lode) Deposits — Gold locked inside rock, placed there by geological processes millions of years ago. Must be drilled, blasted, crushed, and chemically processed.

Secondary (Alluvial / Placer) Deposits — Gold freed from rock by natural weathering and concentrated in rivers by gravity. Extracted using simple gravity methods.

2.2 Major Hard Rock Deposit Types

TYPE	FORMATION	GRADE	SIZE	KEY EXAMPLES
Orogenic	Hot fluids along deep fault zones during mountain-building	1–20 g/t	<0.5–50+ Moz	Obuasi (Ghana), Loulo (Mali)
Epithermal	Near-surface (<1.5 km), linked to volcanic activity	1–15 g/t	0.5–10 Moz	Hishikari (Japan)
Porphyry	Massive, low-grade, around cooling magma bodies	0.3–1.5 g/t	5–100+ Moz	Grasberg (Indonesia)
Carlin-type	Ultra-fine "invisible" gold in carbonate rocks	1–10 g/t	1–30 Moz	Carlin Trend (Nevada)
VMS	Underwater volcanic centres; gold as by-product	Variable	Variable	—

> **The most important type in West Africa is Orogenic** — this is what makes West Africa one of the world's premier gold regions. Nearly all major West African gold mines exploit orogenic deposits hosted in Birimian greenstone belts.

2.3 Alluvial (Placer) Gold Deposits

Alluvial gold forms when primary deposits are weathered and eroded. Gold particles, being 7× heavier than surrounding sand, sink to the bottom of rivers and concentrate at specific points.

Where Gold Concentrates in a River

- **Bedrock surface** — Gold sinks through gravel until it hits solid rock. The bottom 0.5–2.0 m of gravel plus the top 0.3–1.0 m of weathered bedrock is the "pay zone"
- **Bedrock cracks and crevices** — Gold gets trapped in irregularities. Thorough bedrock cleaning is critical
- **Inside of river bends** — Water slows on the inner bank, dropping heavy particles (point bars)
- **Behind obstacles** — Large boulders create low-velocity zones where gold settles
- **Where gradient changes** — Steep stream entering a flat valley; gold drops out as velocity decreases

Types of Secondary Deposits

TYPE	DISTANCE FROM SOURCE	GOLD SHAPE	NOTES
Eluvial (residual)	0–50 m	Angular, rough	Source is directly uphill
Colluvial (slope)	50–500 m	Sub-angular	Transitional
Alluvial (river)	500 m–tens of km	Rounded, flat	Classic placer deposits

2.4 West Africa Gold Geology

West Africa sits on the **West African Craton** — one of Earth's oldest continental cores (>2.0 billion years). The southern **Leo-Man Shield** hosts the **Birimian greenstone belts**, among the most gold-endowed geological formations on Earth.

The key event: the **Eburnean Orogeny** (~2.15–2.07 billion years ago) created deep crustal shear zones that channelled gold-bearing fluids upward, depositing the orogenic gold deposits we mine today.

Major Gold Belts in West Africa

BELT	LOCATION	KEY DEPOSITS	SIGNIFICANCE
Ashanti Belt	Ghana	Obuasi (>30 Moz), Tarkwa	Reference for orogenic gold
Senegalo-Malian Shear Zone	Mali–Senegal	Loulo (~15+ Moz), Fekola (~10+ Moz)	Major alluvial and hard rock gold province
Siguiri Basin	Guinea	Siguiri (>10 Moz)	Major gold-producing region
Ity-Bondoukou Belt	Côte d'Ivoire	Ity (>5 Moz), Yaouré (~3 Moz)	Rapidly growing exploration frontier
Burkina Faso Belts	Burkina Faso	Essakane (>10 Moz)	Regional context

Part 3: Exploration — From Idea to Drill Target ALL BD MGMT

3.1 The Exploration Pipeline

Gold exploration follows a systematic funnel: start wide, narrow down progressively. At each stage, you spend more money but reduce uncertainty. The industry average success rate from grassroots to economic discovery is roughly **1 in 500 to 1 in 1,000**.

STAGE	ACTIVITIES	COST	TIMELINE	KEY DELIVERABLE
1. Desktop Study	Review geological maps, satellite imagery, published data	$50K–$500K	1–6 months	Target areas identified
2. Remote Sensing	Satellite analysis, airborne magnetics, gravity	$100K–$2M	3–12 months	Structural maps, anomaly targets
3. Geological Mapping	Field geologists record rocks, structures, alteration	$200K–$1M	2–6 months	Detailed geological maps
4. Geochemistry	Stream sediment, soil sampling (grid), rock chip	$500K–$5M	6–18 months	Gold-in-soil anomalies
5. Ground Geophysics	IP, EM — detect sulfide mineralization through soil	$500K–$3M	3–12 months	Refined drill targets
6. Initial Drilling	Wide-spaced (100–200 m) RC drilling	$2M–$10M	6–18 months	Discovery confirmation
7. Resource Drilling	Infill RC + DD at 25–50 m spacing, 3D modelling	$10M–$50M	1–3 years	JORC/NI 43-101 Resource
8. Feasibility	PEA → PFS → BFS engineering studies	$2M–$30M	1–3 years	Technical/economic viability

Stage 1: Desktop Study (Literature Review & Target Generation)

The desktop study is where every exploration program begins — and where the best geologists earn their keep before spending a dollar in the field. The goal is to compile and analyze all existing information about a region to identify the most promising areas for follow-up.

What it involves:

- Reviewing published geological maps (typically 1:200,000 or 1:100,000 scale from national geological surveys)
- Compiling historical exploration reports — previous companies may have drilled, sampled, or mapped the area. Their data is often publicly available (or purchasable) and can save millions
- Analyzing academic research papers on regional geology, structural controls, and metallogeny
- Reviewing government databases for known mineral occurrences, artisanal mining sites, and past production records
- Downloading freely available satellite imagery (Landsat, Sentinel-2, ASTER) for preliminary interpretation

Why it matters: A thorough desktop study can eliminate 90% of an area from consideration before any fieldwork begins. The best exploration geologists find deposits that others missed — often because they read reports that were sitting in a government archive for decades. This stage is extremely high-ROI: $50K–$500K spent here can save $5M+ in wasted field programs.

Key deliverables: Target area map, compilation database, summary report ranking targets by priority.

Stage 2: Remote Sensing & Airborne Surveys

Remote sensing bridges the gap between desktop research and boots-on-the-ground fieldwork. It uses satellite and airborne instruments to detect geological features and mineral signatures invisible to the naked eye.

What it involves:

- **Multispectral/Hyperspectral satellite analysis:** Processing satellite data (ASTER, WorldView-3, Sentinel-2) to map alteration minerals associated with gold — iron oxides (gossans), clay minerals (sericite, kaolinite), carbonates. These minerals form "alteration halos" around gold-bearing structures
- **Airborne magnetics:** Flying a magnetometer on a plane or helicopter at 50–80 m above ground to map magnetic minerals in the subsurface. In West Africa, this is critical because thick laterite cover (10–60 m) hides the underlying geology
- **Airborne radiometrics:** Measures natural radioactivity (K, U, Th) from rocks — helps map different rock types and identify potassic alteration associated with some gold systems
- **Digital Elevation Models (DEMs):** Extracting structural lineaments (faults, shear zones) from topographic data — gold deposits in West Africa are overwhelmingly controlled by structures

Why it matters: Airborne magnetic surveys are arguably the single most important geophysical tool in West African gold exploration. The Birimian greenstone belts that host 95%+ of West Africa's gold are strongly magnetic, and structures that control gold mineralization create clear magnetic signatures. A regional aeromagnetic survey can define drill targets that would take years to find through surface methods alone.

Key deliverables: Structural interpretation maps, alteration anomaly maps, prioritized target list for field follow-up.

Stage 3: Geological Mapping (Field Reconnaissance)

This is when geologists finally walk the ground. Geological mapping is the fundamental skill of exploration — everything else builds on it. The geologist's job is to observe, record, and interpret what the rocks are telling them.

What it involves:

- Walking systematic traverses across the target area (typically at 200–400 m spacing for regional mapping, 50–100 m for detailed)
- Recording rock types (lithology), structural features (bedding, foliation, faults, shear zones, quartz veins), and alteration (color changes, new minerals)
- Collecting rock samples for thin section petrography and geochemical analysis
- Mapping artisanal mining sites — in West Africa, local miners are often the best pathfinders. If artisanal miners are digging in an area, there is gold. The question is how much and at what grade
- Recording GPS locations, photographs, and field notes that will be compiled into a geological map

Why it matters: No amount of remote sensing or geophysics can replace a good geologist on the ground. Mapping reveals the 3D geometry of the geology — how rocks are folded, faulted, and intruded — which controls where gold is deposited. A well-mapped project also builds credibility with investors and joint venture partners.

Challenges in West Africa: Deep laterite weathering means rock outcrop is rare (sometimes <5% of the surface). Geologists must rely on road cuts, river beds, artisanal pits, and termite mounds (which bring subsurface material to surface) to see the geology.

Key deliverables: Geological map (1:10,000 to 1:50,000 scale), structural interpretation, sample locations, field report with target recommendations.

Stage 4: Geochemistry (Soil, Stream & Rock Sampling)

Geochemistry is the process of systematically collecting and analyzing earth materials to find chemical signatures of buried gold mineralization. It is the workhorse method for defining drill targets in West Africa.

What it involves:

- **Stream sediment sampling:** Collecting fine-grained sediment (<200 mesh / 75 μm) from active stream beds. One sample integrates the geochemistry of the entire upstream catchment. Gold and pathfinder elements (As, Sb, W, Bi) above background indicate a source somewhere upstream
- **Soil sampling:** The most important surface exploration method in gold exploration. Samples are collected from the B or C soil horizon (typically 30–80 cm depth) on a systematic grid. Typical spacing is 100–200 m between lines and 25–50 m between sample points
- **Rock chip sampling:** Collecting rock fragments from outcrops, float, or artisanal mine workings. Provides direct evidence of gold grades but is limited by exposure
- **Termite mound sampling:** Unique to tropical Africa — termites bring material from 10–40 m depth to surface, providing a "drill hole" through the laterite cap

Why it matters: A well-executed soil sampling program can define gold anomalies that directly correspond to underlying mineralization — even through 20–40 m of barren laterite cover. The key is understanding what you're sampling: in lateritic terrain, gold is often concentrated in specific soil horizons, and sampling the wrong horizon gives false negatives.

Cost context: A regional stream sediment survey (500–1,000 samples at $30–50/sample including collection) costs $30K–$80K and can cover hundreds of km². A detailed soil sampling grid (5,000–20,000 samples) over a priority target costs $300K–$2M. This is one of the highest-value expenditures in exploration.

Key deliverables: Geochemical anomaly maps, gold-in-soil contour plots, target ranking for drill testing.

Stage 5: Ground Geophysics

Ground geophysics provides higher-resolution subsurface information than airborne surveys, helping to precisely locate and define drill targets. At this stage, you're spending real money and need confidence in your targets.

What it involves:

- **Induced Polarization (IP):** Injects electrical current into the ground and measures how rocks store and release charge. Sulfide minerals (pyrite, arsenopyrite) — which are intimately associated with gold in orogenic deposits — produce strong IP anomalies ("chargeability" highs). IP is the most widely used ground geophysical method in gold exploration
- **Electromagnetic (EM) surveys:** Ground EM (moving loop, fixed loop) maps conductive features — graphitic shear zones, sulfide-rich zones, conductive structures. Useful for detecting mineralized structures beneath thick cover
- **Gravity surveys:** Measures subtle variations in gravitational acceleration caused by density contrasts in rocks. Can detect quartz-filled structures, lithological contacts, and faults
- **Ground magnetics:** Higher-resolution version of airborne magnetics, used to map specific structures at target scale

Why it matters: IP surveys are particularly powerful in West Africa because orogenic gold deposits are almost always associated with sulfide minerals. A coincident gold-in-soil anomaly + IP chargeability high + structural target from magnetics = a high-priority drill target. This multi-dataset approach dramatically improves drill success rates.

Typical survey design: IP lines are surveyed at 100–200 m spacing with dipole-dipole or pole-dipole arrays. A single IP line costs $3K–$8K depending on terrain and line length. A typical ground IP program covers 50–200 line-km.

Key deliverables: IP chargeability and resistivity sections, integrated target maps combining geology + geochemistry + geophysics, drill hole location plans.

Stage 6: Initial Drilling (Discovery Drilling)

This is the moment of truth. After all the desktop analysis, remote sensing, mapping, sampling, and geophysical surveys, you finally put steel in the ground to test whether there is gold at economic grades and widths.

What it involves:

- **Reverse Circulation (RC) drilling** at wide spacing (typically 100–200 m between holes, sometimes 400 m for first-pass testing)
- Holes are typically 100–200 m deep in West Africa (deeper if targeting known structures at depth)
- Samples are collected every 1 m and submitted for fire assay
- Geological logging of RC chips to build a subsurface geological model
- Typically 20–60 holes in an initial program, testing 3–8 targets

Why it matters: This is where the vast majority of exploration programs end. Statistically, only about 1 in 10 initial drilling programs find mineralization worth following up, and only about 1 in 100 grassroots targets becomes an economic deposit. A "discovery hole" — one that intersects significant gold mineralization over economic widths — can transform a project overnight. A single result like "48m @ 3.2 g/t gold from 52m depth" can multiply a project's value 10x.

What a good result looks like:

- **Grade x width** is what matters. "2m @ 50 g/t" (narrow, high-grade) and "100m @ 1 g/t" (wide, lower-grade) are both interesting but represent very different deposit types
- Consistent mineralization over meaningful widths (>5m for open-pittable, >2m for underground at high grades)
- Multiple holes intersecting mineralization — a single hole is never enough

What kills a target: Consistently sub-economic grades (<0.5 g/t over narrow widths), erratic gold distribution with no geological continuity, or metallurgical issues (refractory gold that doesn't respond to standard processing).

Key deliverables: Drill results (significant intercepts), cross-sections, geological model, decision to advance or drop the target.

Stage 7: Resource Drilling (Infill & Definition Drilling)

If initial drilling confirms a discovery, the next step is to drill enough holes to define the size, shape, and grade of the deposit so that a Mineral Resource Estimate can be prepared.

What it involves:

- **Infill RC drilling** at 50–100 m spacing to improve geological confidence
- **Diamond Drilling (DD)** at 25–50 m spacing for high-confidence resource categories. DD produces continuous rock core that provides:
  - Detailed geological, structural, and alteration information
  - Oriented core for structural measurements
  - Metallurgical test samples (you need core to determine how to process the ore)
  - Geotechnical data for mine design (rock strength, fracturing)
- **3D geological modelling** using software (Leapfrog, Micromine, Datamine) to build wireframes of ore zones
- **Geostatistical analysis** (variography, kriging) to estimate grade distribution

Why it matters: The Mineral Resource Estimate (MRE) is the single most important document in the life of a mining project. It classifies the deposit into:

- **Inferred Resource:** Low geological confidence (wide-spaced drilling). Good enough to demonstrate potential but not to base investment decisions on
- **Indicated Resource:** Moderate confidence (closer-spaced drilling). Sufficient for a Pre-Feasibility Study
- **Measured Resource:** High confidence (close-spaced drilling). Required for a Definitive Feasibility Study and mine permitting

The MRE must be prepared by a Competent Person (CP) under the JORC Code or a Qualified Person (QP) under NI 43-101. The CP/QP takes personal legal responsibility for the estimate.

Cost context: Resource drilling is the most capital-intensive exploration stage. A typical program involves 30,000–100,000 m of combined RC + DD drilling at $30–200/m, plus geological modelling, metallurgical testing, and the resource estimate itself.

Key deliverables: JORC/NI 43-101 Mineral Resource Estimate, 3D geological model, metallurgical test results, cross-sections and long-sections showing ore zone geometry.

Stage 8: Feasibility Studies (PEA → PFS → DFS)

Feasibility studies are the engineering and economic analyses that determine whether a mineral resource can be profitably mined. They progress through three levels of increasing detail and cost.

Preliminary Economic Assessment (PEA):

- First economic "sanity check" — can this deposit make money?
- Order-of-magnitude engineering: conceptual mine plan, generic processing flowsheet, rough capital and operating cost estimates (±35–50% accuracy)
- Typically costs $500K–$2M
- Can be based on Inferred Resources (unlike PFS/DFS)

Pre-Feasibility Study (PFS):

- More detailed engineering study (±25% cost accuracy)
- Mine design, processing plant design, infrastructure layout, environmental baseline
- Must be based on at least Indicated Resources
- Typically costs $3M–$10M
- Banks and major investors typically require at least a PFS before committing capital

Definitive Feasibility Study (DFS / Bankable Feasibility Study / BFS):

- Full engineering study (±15% cost accuracy) — the basis for a mine construction decision
- Detailed mine plan, process plant engineering, infrastructure design, environmental and social impact assessments
- Must be based on Measured + Indicated Resources
- Includes pilot plant metallurgical testing, geotechnical studies, hydrology, detailed CAPEX/OPEX estimation
- Typically costs $10M–$30M+
- This is the document that banks use to approve project finance (hence "bankable")

Key economic metrics assessed:

- **NPV (Net Present Value):** The discounted present value of future cash flows, minus capital costs. Positive NPV = the project creates value
- **IRR (Internal Rate of Return):** The discount rate at which NPV = 0. Most gold projects need >15% IRR to attract investment, >20% to be considered strong
- **AISC (All-In Sustaining Cost):** Total cost per ounce of gold produced, including mining, processing, G&A, sustaining capital, royalties. Currently, the industry average is ~$1,300–$1,500/oz
- **Payback period:** Time to recover initial capital investment. Banks prefer <5 years

Key deliverables: Technical report (JORC/NI 43-101 compliant), financial model, mine plan, environmental and social impact assessment, risk register, recommendation to proceed (or not) to construction.

Drilling Methods

METHOD	DEPTH	SAMPLE QUALITY	COST	USE
RAB (Rotary Air Blast)	10–50 m	Low	$10–20/m	Reconnaissance, penetrating laterite
RC (Reverse Circulation)	50–300 m	Good	$30–80/m	**Main workhorse for West Africa**
DD (Diamond Drilling)	>2,000 m	Highest (core)	$80–200/m	Detailed geology, metallurgical samples

Fire Assay — The Gold Standard of Analysis

Fire assay has been used for over 500 years to determine gold content:

1. **Sample preparation:** Crush 2–3 kg to <2 mm, split, pulverize to >90% passing 75 µm
2. **Fusion:** 30–50 g sample + flux (PbO, Na_2CO_3, borax) melted at 1,100°C
3. **Cupellation:** Lead button oxidized at 950°C, leaving gold-silver bead
4. **Measurement:** Weighed (gravimetric) or dissolved for AAS/ICP analysis

> **QA/QC is critical:** Every drill program must include blanks (check contamination), certified reference materials (check accuracy), duplicates (check reproducibility), and umpire assays (5–10% sent to a second lab).

The "Nugget Effect"

Gold is distributed extremely unevenly in deposits. A small sample may miss gold entirely, or include one large particle that inflates the result. This is the biggest challenge in estimating alluvial gold grades. **Solution: use large-volume (bulk) samples.**

3.4 Remote Sensing & Satellite Technology MGMT BD

Remote sensing is one of the most powerful and cost-effective tools in modern gold exploration. It allows geologists to analyze vast areas from space before ever setting foot on the ground.

Satellite Platforms Used in Exploration

PLATFORM	RESOLUTION	KEY CAPABILITIES	COST
Landsat 8/9 (NASA)	30 m multispectral, 15 m pan	Iron oxide mapping, vegetation anomalies, regional geology	Free
Sentinel-2 (ESA)	10–20 m multispectral	Similar to Landsat but higher resolution; lithological mapping	Free
ASTER (NASA/Japan)	15–90 m, 14 bands	Best free platform for alteration mineral mapping (clays, iron oxides, carbonates)	Free
WorldView-3 (Maxar)	0.3 m visible, 3.7 m SWIR	Industry-leading mineral discrimination; can identify specific alteration assemblages from space	$15–30/km²
PlanetScope	3 m daily	Very high temporal resolution; monitor site changes, vegetation stress	$5–15/km²
ALOS PALSAR (JAXA)	10–25 m SAR	Radar penetrates clouds and vegetation; structural mapping in tropical West Africa	Free/Low cost

Key Remote Sensing Techniques for Gold Exploration

1. Spectral Analysis — Mapping Alteration Minerals

Gold deposits create "alteration halos" — zones of chemically altered rock around the mineralization. These alteration minerals have distinct spectral signatures detectable from space:

ALTERATION TYPE	MINERALS	SPECTRAL BAND	SIGNIFICANCE
Iron oxide	Hematite, goethite, jarosite	VNIR (0.4–1.0 μm)	Indicates weathered sulfide zones; "gossans" are classic surface indicators of underlying mineralization
Hydroxyl (clay)	Sericite, kaolinite, illite	SWIR (2.0–2.5 μm)	Hydrothermal alteration — the "plumbing system" that carried gold-bearing fluids
Carbonate	Calcite, dolomite, ankerite	SWIR (2.3–2.5 μm)	Carbonate alteration associated with orogenic gold
Silicification	Quartz flooding	TIR (8–12 μm)	Direct host of gold in many orogenic deposits

2. Structural Analysis — Finding the Plumbing

Gold deposits in West Africa are controlled by major crustal structures (shear zones, faults). Remote sensing maps these structures through:

- **Lineament extraction** from digital elevation models (SRTM, ALOS)
- **Magnetic data** (airborne or satellite-derived) to map deep crustal structures beneath laterite cover

- **Radar imagery** (SAR) that penetrates vegetation canopy — critical in tropical West Africa where forest cover hides outcrop

3. Multi-Layer Integration (GIS Overlay)

The most powerful approach combines multiple data layers in a Geographic Information System (GIS):

```
Layer 1: Geological map (rock types, known deposits)
Layer 2: Structural map (faults, shear zones from DEM + magnetics)
Layer 3: Alteration map (iron oxide + clay anomalies from satellite)
Layer 4: Geochemical data (stream sediment, soil sampling results)
Layer 5: Geophysical data (magnetics, IP, gravity)
──────────────────────────────────────────────────
Output: Ranked target areas where multiple layers converge
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> **The "four-layer overlay" is a powerful exploration method** — combining geological mapping, structural interpretation, satellite alteration analysis, and geochemical data to generate drill targets. It is widely used in West African gold exploration.

AI and Machine Learning in Exploration

A growing number of companies are applying AI/ML to mineral exploration:

COMPANY	APPROACH	NOTABLE RESULTS
KoBold Metals	ML-driven target generation for battery metals	$1.2B+ raised; Mingomba discovery (Zambia)
Ivanhoe Electric	Proprietary geophysics + data science	Typhon copper deposit (USA)
Earth AI	ML for drill targeting in brownfield settings	Multiple verified targets in Australia
Goldspot Discoveries	AI analytics for resource companies	Multiple successful predictions

Key applications:

- **Prospectivity mapping:** Train ML models on known deposits, predict undiscovered ones
- **Geochemical prediction:** Use satellite imagery to predict soil geochemistry, reducing the need for expensive field sampling
- **Drill target optimization:** Analyze multi-source data to rank and prioritize drill targets
- **Core logging automation:** AI-assisted identification of rock types and alteration from drill core photos

> **Important caveat:** AI in exploration is a tool that enhances geologist decision-making, not a replacement for geological understanding. The best results come from combining AI analytics with experienced geological interpretation.

Part 4: Resource Estimation & Reporting ALL

LEGAL FINANCE

4.1 Resources vs Reserves

This is one of the most important concepts in mining. The distinction has **legal, financial, and technical implications**.

Mineral Resources — Classified by Confidence

CATEGORY	CONFIDENCE	CAN BE USED FOR
Inferred	Low — geologically reasonable but uncertain	PEA only. **Cannot be used in feasibility studies**
Indicated	Moderate — reasonable assumption of continuity	Pre-Feasibility Study (PFS)
Measured	High — known with high confidence	Feasibility Study (FS/BFS)

Mineral Reserves — The Economically Mineable Part

CATEGORY	DERIVED FROM	CONFIDENCE
Probable Reserve	Indicated Resource	Moderate
Proven Reserve	Measured Resource	High

> **Key principle:** Resources are NOT Reserves. Many published Resources will never become Reserves because they fail economic, environmental, or other tests.

4.2 Reporting Standards

STANDARD	JURISDICTION	EXCHANGE	QUALIFIED PROFESSIONAL
JORC	Australia	ASX	Competent Person (CP)
NI 43-101	Canada	TSX, TSX-V	Qualified Person (QP)
S-K 1300	USA	NYSE, NASDAQ	Qualified Person

> **Legal implications:** You **cannot** publicly claim a "resource" or "reserve" without a report signed by a CP/QP under the relevant standard. Using resource/reserve terminology without qualification can constitute securities fraud. A Competent Person must have at least 5 years' relevant experience.

4.3 The Feasibility Ladder

STUDY	ACCURACY	BASED ON	PURPOSE	COST
PEA	±35–50%	Inferred + Indicated	Worth investigating further?	$0.5–2M
PFS	±25–30%	Min. Indicated	Likely economic? Begin permitting	$2–10M
DFS / BFS	±10–15%	Measured + Indicated	Final investment decision; required by banks	$5–30M

Part 5: Mining & Processing Methods ALL OPS

5.1 Open Pit Mining

The most common method in West Africa, especially for near-surface deposits with thick oxidized zones.

PARAMETER	DESCRIPTION	TYPICAL VALUES
Overall Slope Angle	Angle from pit bottom to rim	Laterite: 35–45°, Fresh rock: 45–60°
Bench Height	Height of each step	Small: 5–6 m, Large: 10–15 m
Strip Ratio	Waste : Ore (tonnes)	Economic: 3:1 to 10:1
Min Pit Bottom Width	Space for equipment	30–40 m minimum

Daily Operations Cycle

1. **Drilling** — Blast holes on a grid (2.5–4.0 m spacing)
2. **Blasting** — ANFO or emulsion explosives; strict safety clearance required
3. **Loading** — Excavator loads trucks; grade control directs ore vs waste
4. **Hauling** — Trucks carry ore to crusher, waste to dump
5. **Dewatering** — Continuous pumping, especially critical during rainy season

5.2 Underground Mining

Used when deposits extend too deep for economic open pit mining. Most common method for West African gold:

- **Cut and Fill** — Mine one horizontal slice, backfill, mine next. Best for narrow, high-grade orogenic veins
- **Sub-Level Stoping** — For steep, regular ore bodies; drill and blast from different levels

5.3 Artisanal & Small-Scale Mining (ASM)

ASM represents one of the oldest and most widespread forms of mining globally, employing an estimated 15–20 million people worldwide. In Africa alone, ASM supports the livelihoods of over 10 million miners and their families.

Defining Characteristics

- Low mechanization — ranging from hand tools (picks, shovels, pans) to small excavators and basic wash plants
- Processing primarily by gravity methods (sluice boxes, panning, simple centrifugal concentrators)
- Labour-intensive, typically employing large numbers of local workers per operation
- Produces approximately 600–800 tonnes of gold globally per year, representing 15–20% of total world mine production
- Often operating in informal or semi-formal regulatory frameworks, with formalization efforts underway in many African countries

Scale Categories

CATEGORY	WORKFORCE	EQUIPMENT	ANNUAL OUTPUT
Artisanal (hand mining)	5–50 workers	Hand tools, gold pans, sluice boxes	0.5–5 kg Au/year
Small-scale mechanized	50–300 workers	Excavator, trommel, wash plant	5–50 kg Au/year
Semi-industrial	100–500+ workers	Multiple excavators, processing plant, generator	50–500 kg Au/year

Key Equipment for Mechanized Alluvial Operations

- Excavator (20–36 tonne class) for digging alluvial gravel
- Trommel screen or vibrating screen to remove oversize material (>20 mm)
- Sluice boxes, jigs, or spiral concentrators for primary gravity concentration
- Shaking tables or centrifugal concentrators (e.g., Knelson, iCON) for final upgrading
- Borax smelting furnace to produce doré bars

ASM and the Global Gold Supply Chain

ASM-produced gold enters the formal supply chain through licensed buying offices, government-approved trading houses, or — in many cases — informal channels. Efforts to formalize ASM supply chains include the LBMA Responsible Gold Guidance, the OECD Due Diligence Guidance, and national certification schemes in countries like the DRC (ITSCI), Colombia, and several West African nations.

Key Challenges

- **Environmental impact** — mercury use (being replaced by borax smelting), deforestation, river siltation

- **Occupational health and safety** — unshored pits, dust exposure, chemical handling without PPE
- **Regulatory complexity** — overlapping claims, contested permits, evolving legal frameworks
- **Community dynamics** — ASM often predates formal exploration and establishes de facto land rights

5.4 Gold Processing — Key Methods

Gold processing transforms raw ore or alluvial material into refined gold. The choice of processing method depends on the ore type (free-milling vs. refractory), gold particle size (coarse vs. fine), grade, and scale of operation. Most operations use a combination of methods in sequence.

Processing Decision Tree

ORE TYPE	GOLD CHARACTERISTICS	RECOMMENDED METHOD	EXPECTED RECOVERY
Alluvial gravel	Coarse, free gold (>100 µm)	Gravity separation	70–90%
Oxidized rock ore	Free gold, moderate grade	Gravity + CIL/CIP	90–96%
Fresh (sulfide) rock ore	Gold locked in sulfides	Flotation + CIL/CIP or roasting	85–95%
Low-grade, large tonnage	Fine gold, oxide ore	Heap leaching	50–75%
Refractory ore	Gold locked in arsenopyrite/pyrite	Oxidation (POX/BIOX/roasting) + CIL	80–95%
Double-refractory	Gold locked in sulfides + carbonaceous (preg-robbing)	Ultra-fine grinding + oxidation + CIL	70–90%

Gravity Separation

Exploits the density difference between gold (19.3 g/cm³) and waste minerals (~2.65 g/cm³). The **oldest and safest** gold recovery method, and the foundation of alluvial gold mining.

How gravity separation works: Gold is approximately 7x denser than the surrounding sand, gravel, and rock. When gold-bearing material is agitated in water, gold particles settle faster and concentrate at the bottom. Every gravity device exploits this principle in different ways.

EQUIPMENT	HOW IT WORKS	RECOVERY	BEST FOR	THROUGHPUT
Gold Pan	Manual swirling in water; tilt to wash away light material	50–70% (operator-dependent)	Prospecting, sampling	0.01 t/hr
Sluice Box	Inclined trough with riffles; gold settles behind riffles, lighter material washes away	60–80% (coarse gold)	ASM, alluvial	5–50 t/hr

EQUIPMENT	HOW IT WORKS	RECOVERY	BEST FOR	THROUGHPUT
Jig	Pulsating water stratifies particles by density in repeated cycles	70–85%	Medium scale	10–100 t/hr
Spiral Concentrator	Slurry flows down a helical trough; centrifugal + gravity separates heavy from light	50–70%	Continuous operations, no power needed	1–5 t/hr per spiral
Shaking Table	Asymmetric reciprocating motion on an inclined deck with riffles separates by density and size	>95% coarse gold	Final cleaning / upgrading of concentrates	0.5–2 t/hr
Centrifugal Concentrator (Knelson/Falcon)	Enhanced gravity field (60–200G) forces fine gold into retention zones	>95% including fine gold (<50 µm)	Industrial and semi-industrial	1–100 t/hr
Trommel	Rotating cylindrical screen; classifies material by size before gravity concentration	N/A (classification only)	Pre-screening alluvial gravel	50–300 t/hr

Gravity Circuit Design — Typical Alluvial Operation:

1. **Excavation:** Excavator digs alluvial gravel from the deposit
2. **Screening:** Trommel or vibrating screen removes oversize boulders and clay (>20 mm material rejected)
3. **Primary concentration:** Sluice box or jig recovers coarse gold (+200 µm)
4. **Secondary concentration:** Spiral or centrifugal concentrator recovers fine gold from sluice tailings
5. **Final upgrading:** Shaking table produces a high-grade concentrate (30–70% gold)
6. **Smelting:** Concentrate is smelted with borax to produce doré

Key operating parameters:

- **Feed rate** must match equipment capacity — overloading reduces recovery
- **Water flow** is critical: too little and light minerals don't wash away; too much and fine gold is lost
- **Gold liberation size** — gravity works best on free gold particles. If gold is locked inside sulfide minerals, gravity alone won't recover it
- **Clean-up frequency** — sluice boxes and jigs need regular clean-up (every 4–12 hours depending on grade) to prevent gold build-up reducing efficiency

Cyanide Leaching (CIL / CIP)

The foundation of industrial gold processing. Used for approximately 80% of global gold production. Cyanide is the only reagent that can economically dissolve fine gold from large volumes of ore.

The Chemistry:

Gold dissolves in dilute cyanide solution in the presence of oxygen — a reaction known as Elsner's equation:

```
4Au + 8NaCN + O2 + 2H2O → 4Na[Au(CN)2] + 4NaOH
```

This reaction is slow (24–72 hours for full dissolution) and requires careful control of pH (10–11, maintained with lime to prevent HCN gas generation), dissolved oxygen, and cyanide concentration.

CIL (Carbon-in-Leach) Process — Step by Step:

1. **Crushing:** Run-of-mine ore is crushed in jaw and cone crushers to <12 mm
2. **Grinding:** Ball mills or SAG mills grind the ore to ~75 μm (200 mesh) — a fine powder where ~80% of gold is liberated from the host rock
3. **Thickening:** Slurry is thickened to 40–50% solids to reduce water consumption
4. **Leaching:** Slurry is pumped through 6–10 large agitated tanks (each 500–2,000 m^3) in series. Sodium cyanide (200–500 ppm) is added, and oxygen is sparged into the slurry
5. **Carbon adsorption:** Activated carbon (coconut shell, 6–12 mesh) is added directly into the leach tanks. The carbon adsorbs dissolved gold from solution — this is the "carbon-in-leach" step
6. **Carbon recovery:** Carbon is screened from the slurry using inter-stage screens and advanced counter-current to the slurry flow
7. **Elution (stripping):** Loaded carbon is treated with hot caustic cyanide solution (Zadra or AARL process) to strip gold back into solution
8. **Electrowinning:** Gold-bearing strip solution passes through electrolytic cells where gold is deposited onto steel wool cathodes
9. **Smelting:** Gold-loaded steel wool is smelted at 1,100°C with fluxes to produce doré bars (typically 85–95% gold)
10. **Carbon regeneration:** Stripped carbon is reactivated in a kiln at 700°C and recycled

CIP (Carbon-in-Pulp) vs. CIL:

- **CIP:** Leaching and carbon adsorption happen in separate stages — ore is first fully leached, then carbon is added. Better for slower-leaching ores
- **CIL:** Leaching and adsorption happen simultaneously. Faster, lower gold inventory in circuit. Preferred for most new plants

Recovery: 90–96% for free-milling (oxide) ores
Capital cost: $200–500M+ for a 2–5 Mtpa CIL plant
Operating cost: $15–30/t ore processed (excluding mining)
Cyanide consumption: 0.5–2.0 kg NaCN per tonne of ore

Cyanide Safety:

- Sodium cyanide (NaCN) is highly toxic — lethal dose is 1–3 mg/kg body weight
- Operations must comply with the International Cyanide Management Code (ICMC)
- At pH <9.5, cyanide converts to hydrogen cyanide gas (HCN) — extremely dangerous. This is why pH is maintained above 10 with lime
- Cyanide in tailings is destroyed by oxidation (SO_2/air process, Caro's acid, or hydrogen peroxide) before discharge to the tailings storage facility

Flotation

Flotation is used when gold is finely disseminated within sulfide minerals (pyrite, arsenopyrite) and cannot be liberated by grinding alone. It is a pre-concentration step that produces a sulfide concentrate for further processing.

How it works:

1. Finely ground ore (~75 μm) is mixed with water and chemical reagents in flotation cells
2. **Collectors** (xanthates) make sulfide mineral surfaces hydrophobic (water-repelling)
3. **Frothers** (MIBC, pine oil) create a stable froth on the surface
4. Air is blown through the cell — sulfide minerals (with attached gold) attach to air bubbles and float to the surface
5. The froth (containing the concentrate) is skimmed off; the waste (tailings) sinks

Typical results: Flotation concentrates 90%+ of the gold into 5–15% of the original ore mass, producing a concentrate grading 20–100 g/t gold (compared to 1–5 g/t head grade). This concentrate is then processed by CIL, roasting, or pressure oxidation.

Heap Leaching

A lower-cost alternative for low-grade (0.3–1.5 g/t), large-tonnage, oxidized ores. Heap leaching sacrifices recovery for dramatically lower capital costs.

Process — Step by Step:

1. **Crushing:** Ore is crushed to -19 to -25 mm (coarser than CIL grinding)
2. **Agglomeration:** Fine ore is mixed with cement (5–10 kg/t) to prevent blinding of the heap with fines
3. **Stacking:** Ore is stacked by truck or conveyor/stacker onto an impermeable HDPE liner, typically in 6–10 m lifts
4. **Irrigation:** Dilute cyanide solution (200–500 ppm NaCN) is applied via drip emitters at 8–12 L/hr/m²
5. **Percolation:** Solution percolates down through the heap over 60–120 days, dissolving gold
6. **Collection:** Gold-bearing "pregnant solution" drains to a collection pond at the base of the heap
7. **Gold recovery:** Pregnant solution is processed through activated carbon columns (carbon-in-column, CIC) or Merrill-Crowe zinc precipitation
8. **Recirculation:** Barren solution is re-fortified with cyanide and recirculated

Recovery: 50–75% (lower than CIL but at much lower capital cost)
Capital cost: $50–150M for a 5–20 Mtpa heap leach operation
Operating cost: $5–12/t ore (significantly lower than CIL)
Best suited for: Arid climates (rain dilutes solution), oxide ores (sulfide ores don't leach well), large deposits with low grades

Advantages over CIL:

- 50–70% lower capital cost
- Simpler operation — fewer moving parts, lower technical skill requirements
- Scalable — can start small and expand
- No grinding circuit (the single most expensive part of a CIL plant)

Refractory Ore Treatment

Some gold ores are "refractory" — the gold is physically locked inside sulfide mineral crystals (mainly arsenopyrite and pyrite) and cannot be dissolved by cyanide without first breaking down the sulfide host. Refractory ores require an oxidation step before cyanide leaching.

METHOD	HOW IT WORKS	CAPITAL COST	OPERATING COST	BEST FOR
Roasting	Heat concentrate to 500–700°C in air; sulfides oxidize to iron oxide, releasing gold	High ($300M+)	Moderate	High-grade concentrates, established technology
POX (Pressure Oxidation)	Autoclave at 200°C, 3,000 kPa with oxygen; sulfides dissolve in sulfuric acid	Very high ($400M+)	High	Arsenopyrite-rich ores, best recovery
BIOX (Biological Oxidation)	Bacteria (Acidithiobacillus) eat sulfides over 4–6 days in stirred reactors	Moderate ($200M+)	Low	Lower-grade concentrates, lower CAPEX than POX
Ultra-fine grinding	Grind to <10 µm to physically liberate gold from sulfide crystals	Moderate	Moderate-high (energy)	Moderately refractory ores

Smelting: From Concentrate to Doré

Smelting is the final step at the mine site — converting gold concentrate into a semi-pure gold bar (doré) for shipment to a refinery.

Borax Smelting — the most important method for alluvial and small-scale operations:

1. Mix gravity concentrate (1–30% gold) with borax ($Na_2B_4O_7$) and soda ash (Na_2CO_3) as fluxes
2. Heat to 1,100–1,200°C in a graphite or clay crucible
3. Borax melts into a glassy slag that absorbs silicate and oxide impurities
4. Gold (being densest) sinks to the bottom of the crucible
5. Cool, break slag, extract gold button or bar

Flux recipes vary by concentrate composition:

CONCENTRATE TYPE	BORAX	SODA ASH	SILICA	OTHER
Clean gravity concentrate (>30% Au)	2:1 flux:concentrate	—	—	Minimal flux needed
Iron-rich concentrate	3:1	10%	—	Add manganese dioxide as oxidant
Sulfide-rich concentrate	3:1	15%	10%	Add niter (KNO_3) to oxidize sulfides
Silica-rich concentrate	2:1	20%	—	Extra soda ash to flux silica

Output purity: 85–95% Au (doré), which must be sent to a refinery for further purification to 99.5%+ (London Good Delivery standard) or 99.99% (investment grade).

Industrial smelting at large mines uses induction furnaces (50–500 kg capacity) that can process large batches of electrowinning sludge or gravity concentrate in 2–4 hours. Doré bars are typically 20–30 kg and are shipped under armed escort to a licensed refinery.

Borax smelting is the preferred alternative to mercury amalgamation — non-toxic, higher recovery (90%+ vs mercury's 30–60%), and extremely low equipment cost ($100–1,000). The global movement to eliminate mercury from ASM (Minamata Convention) is driving rapid adoption of borax methods.

Processing Method Comparison Summary

METHOD	CAPITAL COST	OPERATING COST	RECOVERY	MIN. GRADE	SCALE
Gravity only	$0.1–2M	$5–15/t	60–85%	0.5 g/t alluvial	ASM to small mine
Gravity + CIL	$200–500M	$20–40/t	90–96%	1.0 g/t rock	Industrial
Heap leach	$50–150M	$5–12/t	50–75%	0.3 g/t rock	Large, low-grade
Flotation + CIL	$300–600M	$25–45/t	85–95%	1.5 g/t rock	Industrial, sulfide ores
Flotation + POX + CIL	$500M+	$35–55/t	85–95%	2.0 g/t rock	Large, refractory

5.5 Gold Doré

What it is: A semi-pure gold-silver alloy bar produced at the mine through primary smelting. It is the standard product that leaves a mine site.

Doré Purity by Source

SOURCE TYPE	TYPICAL PURITY
Major hard rock mines	90–95% Au
Mid-tier hard rock mines	75–90% Au
Alluvial mines	80–92% Au
ASM hand smelting	70–90% Au

5.6 Health, Safety & Security ALL OPS

Mining is one of the most hazardous industries globally. Every mining team member must understand and prioritize safety.

Common Hazards in Gold Mining

HAZARD CATEGORY	SPECIFIC RISKS	PREVENTION
Ground failure	Pit wall collapse, underground roof fall, trench cave-in	Proper slope design, ground support, no entry to unsupported areas
Mobile equipment	Collision, rollover, struck-by incidents	Traffic management plan, spotters, seatbelts, training
Drowning	Flooded pits, water-filled trenches, river operations	Dewatering, barriers, life jackets for alluvial operations
Falls	Working at height on equipment, steep pit walls	Fall protection, ladders, access controls
Heat stress	West Africa temperatures regularly exceed 40°C	Hydration protocols, shade, work-rest cycles, early starts
Dust	Silicosis from prolonged quartz dust exposure	Wet drilling, dust suppression, respiratory protection
Chemical	Cyanide, fuel, mercury (in ASM areas)	SDS availability, training, PPE, proper storage, cyanide code
Biological	Malaria, waterborne diseases, snake/scorpion bites	Prophylaxis, treated nets, first aid kits, medical evacuation plan
Explosive	Misfired charges, improper storage	Licensed blasters only, strict procedures, magazine security

Minimum PPE Requirements

SITE TYPE	REQUIRED PPE
All mine sites	Hard hat, safety boots (steel toe), high-visibility vest, safety glasses
Drilling operations	+ hearing protection, dust mask/respirator
Processing plant	+ chemical-resistant gloves, face shield (near chemicals)
Alluvial/river operations	+ life jacket, waterproof boots
Laboratory (fire assay)	+ heat-resistant gloves, full face shield, leather apron

Emergency Response

Every mine site must have:

1. **Emergency response plan** — documented, trained, and rehearsed
2. **First aid** — trained first aiders on every shift, stocked first aid stations
3. **Medical evacuation** — defined evacuation route and vehicle, nearest hospital identified, satellite phone or reliable communication
4. **Fire response** — fire extinguishers at all fuel storage, processing, and camp areas

5. **Incident reporting** — all incidents (including near-misses) must be reported and investigated

Security in West Africa

Mining operations in West Africa face specific security challenges:

- **Gold theft** — internal and external. Requires: gold room security, dual custody for all gold handling, CCTV, controlled access, random searches
- **Armed robbery** — particularly during gold transport. Requires: armed escort for gold movements, varied routes and schedules, secure vehicles
- **Community conflict** — disputes over land, employment, or environmental impact. Requires: proactive community engagement (see Section 7.4)
- **Artisanal miner encroachment** — illegal miners on licensed ground. Requires: boundary demarcation, regular patrols, community agreements, government liaison
- **Political instability** — particularly relevant in Mali and Guinea. Requires: evacuation plans, embassy registration, situation monitoring, cash reserves

> **Best practice:** Zero tolerance for safety shortcuts. No production target justifies putting a team member at risk.

5.7 Water Management OPS LEGAL

Water is both essential for mining operations and a critical environmental concern. In West Africa, water management is complicated by extreme seasonal variation — months of drought followed by intense rainy seasons.

Water Uses in Mining

OPERATION	WATER USE	TYPICAL CONSUMPTION
Drilling (RC/DD)	Cooling, dust suppression, sample recovery	5,000–20,000 L/day per rig
Processing (gravity)	Sluice boxes, jigs, wash plants	50–200 m^3/hour
Processing (CIL)	Slurry makeup, reagent mixing	500–2,000 m^3/hour
Dust suppression	Haul roads, stockpiles	Variable, 10,000–50,000 L/day
Camp and domestic	Drinking, cooking, sanitation	100–200 L/person/day

Water Sources

- **Surface water** (rivers, streams) — most common in West Africa but seasonally variable and requires permits
- **Groundwater** (boreholes) — more reliable but may require dewatering permits and impact assessment
- **Pit dewatering** — water pumped from mine pits can often be reused in processing
- **Recycled process water** — modern operations aim to recycle 80–95% of process water

Seasonal Challenges in West Africa

SEASON	CHALLENGE	MANAGEMENT
Dry season (Nov–Apr)	Water scarcity for processing; dust; reduced river flow	Water storage, boreholes, increased recycling
Rainy season (May–Oct)	Pit flooding, road washout, tailings dam overflow risk, increased malaria	Dewatering capacity, drainage infrastructure, road maintenance, stormwater diversions
Transition periods	Flash flooding risk, unstable pit walls from saturation	Monitoring, restricted access, slope stability assessment

Environmental Water Requirements

- **Baseline water quality monitoring** — must be established before mining starts
- **Discharge permits** — any water discharged must meet national standards (typically pH 6–9, TSS <50 mg/L, cyanide <0.2 mg/L)
- **Tailings water management** — zero-discharge is the best practice target; tailings supernatant must be recycled
- **Downstream impact assessment** — monitor water quality downstream of operations
- **Closure obligation** — water quality must be maintained long after mining ceases

> **For alluvial operations:** Water management is the single biggest operational variable. Too little water stops processing; too much floods the site. Successful alluvial operations maintain a reliable water supply year-round through a combination of river abstraction, settling ponds, and recycling.

Part 6: Mine Economics & Valuation FINANCE BD MGMT

6.1 Cost Structure

METRIC	DEFINITION	INCLUDES
C1 Cash Cost	Direct cost per ounce	Mining + Processing + Site G&A + Refining + Royalties
AISC	All-In Sustaining Cost	C1 + Sustaining Capex + Corporate G&A + Sustaining Exploration
AIC	All-In Cost	AISC + Growth Capex + Non-sustaining Exploration

> **Cost advantage:** West African alluvial operations can achieve AISC of **$800–1,000/oz**, placing them in the **lower quartile** of the global cost curve (median ~$1,400/oz). This means alluvial producers can generate strong margins at any gold price.

6.2 Revenue Formula

```
Revenue = Ore processed (t) × Head grade (g/t) × Recovery (%) × Gold price ($/oz) ÷ 31.103
```

6.3 Key Financial Metrics

METRIC	WHAT IT MEANS	MINIMUM THRESHOLD
NPV (Net Present Value)	Present value of all future cash flows minus investment. NPV > 0 = project creates value	Positive, at conservative gold price
IRR (Internal Rate of Return)	Discount rate at which NPV = 0	15–25% (higher for risky jurisdictions)

METRIC	WHAT IT MEANS	MINIMUM THRESHOLD
Payback Period	Time to recover initial investment	Banks want ≤3–5 years

6.4 Valuation Methods

METHOD	BEST FOR	HOW IT WORKS
DCF / NPV	Development to production	Discount all future cash flows to present
EV / Resource oz	Exploration to development	Enterprise Value ÷ Total ounces
Comparable Transactions	Any stage	What did similar projects trade for?
P/NAV	Public companies	Share price ÷ Net Asset Value per share

Typical Valuation Ranges by Stage

STAGE	EV PER RESOURCE OUNCE	P/NAV MULTIPLE
Grassroots exploration	$5–20/oz	—
Advanced exploration	$20–150/oz	0.1–0.3×
Development (feasibility complete)	$100–250/oz	0.3–0.7×
Production	$150–400/oz	0.8–1.2×

> **Key insight for BD:** Value increases dramatically at each de-risking milestone. A JORC resource is worth many times more than drill intercepts. This is the fundamental value creation mechanism in exploration.

6.5 AISC Calculation Example

```
Monthly production:                 2,000 oz Au

Mining cost:                        $320,000
Processing cost:                    $375,000
Site G&A:                           $180,000
Royalty (5% × revenue):             $200,000
Refining fees:                       $10,000
──────────────────────────────────────────────
Total C1 Cash Cost:               $1,085,000  →  $542/oz

Sustaining capex:                   $120,000
Sustaining exploration:              $50,000
──────────────────────────────────────────────
Total AISC:                       $1,255,000  →  $628/oz

At $2,950/oz gold price:
Margin per ounce:                    $2,322
Monthly pre-tax profit:           $4,644,000
```

6.6 Royalties by Country

COUNTRY	ROYALTY	GOV'T FREE CARRY	CORPORATE TAX
Mali	3%	10% + option 10%	25%
Guinea	5%	15%	25%
Cameroon	4%	10%	30%
CDI	3–5% (sliding)	10%	25%
Namibia	3–5%	—	32%
Botswana	3%	15% option	22%

Part 7: Legal & Regulatory Framework LEGAL

MGMT

7.1 Mining Licence Types

In most African jurisdictions, minerals belong to the state. Companies must obtain licences to explore and mine.

LICENCE TYPE	ACTIVITIES	AREA	DURATION
Reconnaissance	Non-invasive surveys	Large (1000s km²)	1–2 years
Exploration Permit	Surveys, sampling, drilling	100–1,000 km²	2–4 years (renewable)
Mining / Exploitation Permit	Full-scale mining	Defined by resource	15–25 years (renewable)

Typical Government Requirements

- **Work commitments:** Minimum exploration expenditure per year
- **Reporting:** Annual technical reports, production data, employment data
- **Environmental:** EIA/ESIA required before mining
- **Local content:** Local hiring and procurement preferences
- **Area relinquishment:** Must give back 25–50% of area at each renewal

7.2 Royalty Structures

TYPE	BASIS	COMMON RATES
Ad Valorem / GRR	% of gross revenue	3–6%
NSR (Net Smelter Return)	% of revenue after smelting/refining costs	1–5%
NPI (Net Profit Interest)	% of net profit after all costs	5–20%

TYPE	BASIS	COMMON RATES
Government Free Carry	Free equity stake in mining company	10–20%

7.3 Joint Venture & Earn-in Agreements

The most common deal structures in exploration-stage mining:

> **Earn-in structure:** Company A (farmer) holds a licence. Company B (farmee) commits to spend $X on exploration over Y years. Upon meeting the expenditure, Company B earns a specified interest (e.g., 51–80%).

Key Contractual Terms

TERM	DEFINITION	WHY IT MATTERS
NSR Royalty	Ongoing royalty on production to original holder	Significantly reduces project economics. Negotiate caps/buybacks
Dilution	What happens if a party doesn't fund its share	Must be precisely defined with formula
Change of Control	What happens if a party is acquired	Can trigger consent or pre-emptive rights
Operatorship	Who manages day-to-day operations	Operator has significant control — ensure approval thresholds
Anti-dilution	Protection below a threshold (usually 10%)	Party typically converts to a royalty
Tail Period	Rights retained after termination	Creates long-term obligations — negotiate carefully

7.4 Environmental & Social Requirements

EIA / ESIA

Required before any mining permit is granted. Includes: baseline environmental studies (12–24 months), impact assessment, mitigation plans, closure plan, public consultation, and independent review.

Social License to Operate

1. **Early, continuous communication** with village chiefs and local government
2. **Prioritize local hiring** — the most effective social licence tool
3. **Fair land compensation** with independent assessors
4. **Community investment:** 1–2% of revenue to community development
5. **Establish a grievance mechanism**

7.5 Country Notes

COUNTRY	KEY POINTS
CDI	Mining Code 2014 (amended 2023). SARL recommended. EP valid 4 yrs, renewable 2×. 10% free carry. Strong French-language legal system
Namibia	Max 4 exploration permits per company. UBO disclosure required. Well-established, investor-friendly framework
Botswana	15% carried interest option. Strong rule of law, one of Africa's most stable
Mali	Code 2019 (amended 2023). Gov't minimum raised to 20%. Royalty 3%. Export permits mandatory. Political risk remains elevated

Part 8: The Gold Value Chain & Market BD

FINANCE

8.1 From Mine to Market

```
Mine Site → On-site Smelting → Doré Bar (80–95% Au)
                                ↓
Refinery (LBMA-certified) → Bullion Bar (≥99.5% Au)
                                ↓
Markets: LBMA London | COMEX | Central Banks | Jewellery | Investment
```

8.2 LBMA Good Delivery Standard

SPECIFICATION	REQUIREMENT
Minimum fineness	995.0 (99.5% pure)
Weight range	350–430 troy ounces (~10.9–13.4 kg)
Marks	Serial number, refiner's stamp, fineness, year

Only ~70 refineries worldwide hold LBMA accreditation. Gold from these facilities is accepted universally. Non-LBMA gold trades at a discount.

8.3 Dubai as a Gold Trading Hub

Dubai (DMCC) is the world's **second-largest physical gold trading hub** after Switzerland.

- 2024 precious metals trade: AED 625 billion (~USD 170 billion)
- 2,500+ precious metals companies in DMCC

- Multiple LBMA-accredited refineries (Kaloti, Emirates Gold, Al Etihad)
- Strategic position between African supply and Asian demand

> **Relevance for small producers:** A DMCC trading entity serves as the entry point for gold into the formal market. Small producers (<1 t/year) typically start with mid-size Dubai traders, then graduate to direct LBMA refinery relationships as compliance documentation matures.

8.4 Gold Pricing

BENCHMARK	DESCRIPTION
LBMA Gold Price	Set twice daily (10:30 AM, 3:00 PM London). The global reference
COMEX Futures	Most actively traded gold futures (CME Group)
Spot Price	Real-time traded price on global markets

8.5 Supply Chain Compliance

LBMA Responsible Gold Guidance (RGG)

All LBMA refineries must conduct due diligence on gold supply chains, based on the **OECD Five-Step Framework**:

1. Establish strong company management systems
2. Identify and assess supply chain risks
3. Design strategy to respond to risks
4. Independent third-party audit
5. Report on due diligence

What Small-Scale Producers Need

- **Chain of Custody** documentation (mine to buyer)
- **KYC** documentation for all counterparties
- **Certificate of Origin** for all gold exports
- Export permits from host country
- Internal compliance policy aligned with OECD framework

Part 9: Industry Landscape BD MGMT

9.1 Industry Tiers

TIER	MARKET CAP	PRODUCTION	EXAMPLES
Major	>$10B	>1 Moz/year	Newmont, Barrick, Agnico Eagle
Mid-Tier	$2–10B	300K–1M oz	B2Gold, Endeavour, Kinross
Junior Producer	$500M–2B	100K–300K oz	Perseus Mining, Galiano Gold
Junior Explorer	<$500M	Pre-production	—
Private / ASM	—	Variable	**Most junior miners start here**

9.2 Key Players in West Africa

COMPANY	HQ	KEY W. AFRICA ASSETS	PRODUCTION
Barrick Gold	Canada	Loulo-Gounkoto (Mali)	~600K oz/yr from Mali
Newmont	USA	Ahafo, Akyem (Ghana)	~1M oz/yr from Ghana
B2Gold	Canada	Fekola (Mali)	~600K oz/yr
Endeavour Mining	UK	Ity, Lafigué (CDI)	~1.4M oz/yr total
AngloGold Ashanti	South Africa	Siguiri (Guinea)	~3M oz/yr total
Perseus Mining	Australia	Yaouré (CDI)	~500K oz/yr

9.3 Junior Mining Lifecycle

```
Grassroots → Discovery → Resource Definition → Feasibility → Development → Production

Most juniors EXIT via acquisition by a larger company at the development stage,
rather than building and operating the mine themselves.
```

9.4 Funding Sources

SOURCE	STAGE	SIZE	DILUTION
Founder's capital	Grassroots	$0.5–5M	None
Angel / Seed	Early exploration	$1–5M	High
IPO (TSX-V/ASX/AIM)	Resource definition	$5–50M	Moderate-High
Streaming	Development	$50–500M	Non-dilutive (but gives up revenue)
Royalty financing	Development	$10–100M	Non-dilutive (permanent cost)
Project finance	Construction (needs BFS)	$100M–1B+	Non-dilutive (debt)
JV / Earn-in	Exploration	$2–20M expenditure	Gives up project equity

9.5 Market Context (2025–2026)

- **Gold price:** Near all-time highs (~$2,950/oz), driven by central bank buying and geopolitical uncertainty
- **Exploration budgets:** Record levels in 2025, West Africa a primary destination
- **M&A activity:** Majors actively acquiring advanced projects and juniors
- **AI in exploration:** KoBold Metals ($1.2B+ raised) deploying ML for target generation
- **Regulatory tightening:** Mali, Guinea, DRC increasing government stakes and export controls

9.6 Technology & Innovation in Mining MGMT BD

The mining industry is undergoing significant technological transformation. Understanding these trends is important for competitive positioning.

Drone Technology

APPLICATION	HOW IT WORKS	COST SAVING VS TRADITIONAL
Topographic survey	Drone-mounted LiDAR or photogrammetry creates detailed 3D terrain models	70–90% cheaper than ground survey for same area
Pit volume measurement	Weekly drone flights to track material movement and reconcile with production data	Replaces manual survey teams

APPLICATION	HOW IT WORKS	COST SAVING VS TRADITIONAL
Exploration mapping	High-resolution orthophotos for geological mapping in areas with poor satellite coverage	Fills gap between satellite (30m) and field mapping
Stockpile measurement	Accurate volume calculations of ore and waste stockpiles	Replaces estimate-based inventory
Environmental monitoring	Track rehabilitation progress, detect illegal mining, monitor water bodies	Covers large areas quickly
Security	Perimeter surveillance, monitoring artisanal miner activity	Supplements ground patrols

Opportunity for junior explorers: Drone surveys for topographic mapping and pit planning at alluvial sites can be done with consumer-grade drones ($2–5K) and free photogrammetry software (OpenDroneMap), delivering professional-grade results at minimal cost.

Portable XRF Analysers

Handheld X-ray fluorescence (XRF) analysers provide instant multi-element geochemistry in the field:

- **Analysis time:** 30–120 seconds per sample
- **Elements detected:** 30+ elements simultaneously (including gold pathfinder elements: As, Cu, Sb, Bi, W)
- **Accuracy:** Semi-quantitative (±10–20% for most elements); not a substitute for fire assay but excellent for screening
- **Cost:** $25–50K per unit
- **Key use:** Real-time decision-making during drilling — determine if an interval is mineralized before waiting weeks for lab results

Limitation: XRF cannot directly detect gold at exploration-relevant grades (<10 g/t) because gold's detection limit by XRF is typically 5–10 ppm. Instead, geologists use **pathfinder elements** (arsenic, antimony, bismuth) that are detectable by XRF and commonly associated with orogenic gold.

Data Management & Digital Geology

Modern exploration generates enormous amounts of data. Key tools:

TOOL CATEGORY	EXAMPLES	PURPOSE
GIS	QGIS (free), ArcGIS Pro	Spatial data integration, map production, multi-layer analysis
3D Geological Modelling	Leapfrog Geo, Micromine	Build 3D geological models from drill data, generate resource estimates
Database	acQuire, Geobank, DataShed	Centralized storage of all exploration data (drill holes, samples, assays)

TOOL CATEGORY	EXAMPLES	PURPOSE
Remote Sensing	ENVI, Google Earth Engine (free)	Satellite image processing, spectral analysis, change detection
Project Management	Geotrack, custom dashboards	Track exploration expenditure, licence commitments, reporting deadlines

Best practice: Building a centralized geological database early is critical. As operations scale across multiple countries, the ability to query, compare, and analyze data across all projects becomes a major competitive advantage.

Emerging Technologies to Watch

TECHNOLOGY	MATURITY	POTENTIAL IMPACT
Hyperspectral core scanning	Commercial	Automated mineral identification from drill core; 10x faster than manual logging
Ambient noise tomography	Early commercial	Passive seismic imaging of subsurface structures without active sources
Distributed Acoustic Sensing (DAS)	Research	Converts fibre optic cables into thousands of seismic sensors
Autonomous drilling	Commercial (surface)	Removes operators from hazardous environments; 24/7 operation
Blockchain traceability	Early adoption	End-to-end gold supply chain tracking from mine to market; supports LBMA compliance
Satellite InSAR	Commercial	Monitors ground deformation at mm-scale; essential for tailings dam safety

Part 10: Glossary ALL

TERM	DEFINITION
AISC	All-In Sustaining Cost — total cost per ounce to sustain production
Alluvial	Deposited by water; alluvial gold is found in river sediments and gravel
ASM	Artisanal and Small-Scale Mining
Assay	Laboratory analysis to determine metal content
BFS / DFS	Bankable/Definitive Feasibility Study — final investment decision level
Birimian	Ancient (2.0–2.2 Ga) volcanic-sedimentary rocks in West Africa hosting most gold
Bullion	Refined gold bars meeting LBMA standards (≥99.5%)
C1 Cash Cost	Direct cash cost per ounce (mining + processing + G&A + refining + royalties)
CIL / CIP	Carbon-in-Leach / Carbon-in-Pulp — industrial gold processing methods
CP / QP	Competent Person / Qualified Person — authorized to sign resource reports
Cut-off Grade	Minimum grade for ore to be economic; below this = waste
DCF	Discounted Cash Flow — valuation method
DD	Diamond Drilling — produces rock core, highest quality samples
Dilution	(Mining) Waste mixed with ore. (Corporate) Ownership reduction from new shares
Doré	Semi-pure gold bar (80–95%) produced at mine; needs refining
Earn-in	Earning a project stake by funding exploration expenditure
EIA / ESIA	Environmental (and Social) Impact Assessment
Fire Assay	Standard gold analysis method (fusion at 1,100°C)
Free Carry	Government's free equity in a mining company (10–20%)
Free Gold	Gold recoverable by gravity (not locked in sulfides)
g/t	Grams per tonne — standard hard rock grade unit
g/m³	Grams per cubic metre — standard alluvial grade unit
Greenstone Belt	Belt of metamorphosed volcanic-sedimentary rocks hosting gold
Head Grade	Average gold grade of ore fed into the processing plant
Heap Leach	Low-cost processing: cyanide percolates through crushed ore pile
Inferred Resource	Low-confidence mineral resource; usable only for PEA
Indicated Resource	Moderate-confidence mineral resource; usable for PFS
IP	Induced Polarization — geophysical method detecting sulfides

TERM	DEFINITION
IRR	Internal Rate of Return
JORC	Australian reporting standard for mineral resources/reserves
JV	Joint Venture
Laterite	Deep tropical weathering product covering West African bedrock
LBMA	London Bullion Market Association — sets global gold standard
Measured Resource	High-confidence mineral resource; usable for FS/BFS
Moz	Million ounces
NI 43-101	Canadian reporting standard for mineral resources/reserves
NPV	Net Present Value
NSR	Net Smelter Return royalty
Nugget Effect	Statistical challenge from gold's uneven distribution
Orogenic	Related to mountain-building; main gold deposit type in West Africa
Pay Zone	Layer in alluvial deposit with economic gold grades
PEA	Preliminary Economic Assessment
PFS	Pre-Feasibility Study
Placer	Sand/gravel deposit containing valuable minerals concentrated by gravity
Probable Reserve	Economically mineable part of an Indicated Resource
Proven Reserve	Economically mineable part of a Measured Resource
RC	Reverse Circulation drilling — main workhorse for W. Africa exploration
Recovery Rate	% of gold in ore actually extracted by processing
Refractory Gold	Gold locked inside sulfide minerals; needs pre-treatment
Royalty	Payment to government/licence holder based on production
Shear Zone	Zone of intense crustal deformation; pathway for gold fluids
Strip Ratio	Waste rock : ore ratio
Streaming	Financing: investor buys right to future production at fixed discount
Tailings	Waste material after ore processing
Troy Ounce	31.1035 grams — standard precious metals unit
UBO	Ultimate Beneficial Owner

Appendix: Key Reference Numbers

Current Market Reference (March 2026)

METRIC	VALUE
Gold price (LBMA)	~$2,950/oz
Global AISC median	~$1,400/oz
West Africa alluvial AISC (typical)	$800–1,000/oz
Global mine production (2024)	3,661 tonnes
Mali total production (2024)	58.7 tonnes

Unit Conversions

CONVERSION	VALUE
1 troy ounce	31.1035 grams
1 metric tonne	1,000 kilograms
1 short ton	907.185 kilograms
1 oz/t (per short ton)	34.29 g/t
1 kilogram gold	32.15 troy ounces

This guide is prepared for internal training purposes.
It is designed to give every team member a solid foundation in gold mining and exploration.